

WEEK 01

From a biological question to defensible evidence

Can we explain where every number came from?

By the end, you should be able to...



Match a biological question to an assay and a data type.



Run and save a small notebook without installing software.



Distinguish a prediction, an observation and a justified claim.

Reconnect with the biology

Recall: genes can produce RNA or protein products; RNA counts are not direct protein measurements.

Think silently -> explain to a neighbour -> check the measurement.

Start with the question, not the software

01

Question: does a treatment change the RNA abundance of selected genes?

>

02

Measurement: RNA sequencing of independent biological samples.

>

03

Claim limit: this experiment does not directly measure protein activity.

One case will return all semester

We study a fictional cell-culture response to a treatment.

Tutorial data are synthetic unless the file says REAL EXCERPT.

Known simulation truth lets us test an analysis; it is not a research discovery.

Evidence has a chain of custody

01

Biological sample ->
laboratory assay -> raw
file.

>

02

Raw file -> processing
steps -> table / figure.

>

03

Table / figure ->
interpretation, with
limitations.

Gene counts are versioned answers

19,442

Release 50; stated counting rule

GENCODE release 50 reports 19,442 protein-coding genes under its stated main-chromosome counting rule.

Changing the annotation release or counting definition can change the answer.

Record the release and definition; do not memorize one eternal total.

Notebook = cells + a running memory

01

Text cells explain; code cells calculate.

>

02

The runtime remembers variables in execution order.

>

03

Restart and run all checks whether the notebook works from a clean start.

Our scientific loop

01

Predict what you expect before running the example.

>

02

Run, compare with a control, and inspect a possible defect.

>

03

Explain one result and one thing the result cannot establish.

Reproducible is not automatically correct

Reproducible

Scientifically justified

A fixed seed can reproduce a synthetic dataset.

The same wrong method can reproduce the same wrong answer.

A receipt or checksum checks consistency, not honesty or scientific truth.

AI is optional; responsibility is not

You may ask for explanations or help reading an error where the task permits it.

Record the tool, use and checks; no AI use is also a valid disclosure.

Never paste patient data or classmates' submissions into a public service.

Worked example: a traceable GC fraction

Toy DNA sequence: ACGTGC. Ignore no positions because all six bases are known.

Predict the result and one limitation before calculating.

Work through the reasoning

There are four G or C bases in six positions.

GC fraction = $4 / 6 = 0.667$, approximately 66.7%.

The sequence is synthetic; GC content alone does not identify a species.

Pause: make a defensible claim

The same notebook gives 66.7% to two students. Is that proof they copied? What evidence would you inspect instead?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

No. Identical answers can be correct, and small datasets can collide. Inspect the submitted work and reasoning; discuss concerns fairly rather than infer misconduct from a number.

Today's practical

W01: run the setup, calculate GC on your assigned toy sequence, record a prediction and save a notebook plus a summary JSON.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 assay
- 2 raw data
- 3 notebook
- 4 runtime
- 5 seed

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: change the sequence by one base and predict how GC changes before running it.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S01 Alberts et al. Molecular Biology of the Cell, 4th ed. Foundational cell biology chapters.

S02 GENCODE Human Release 50 and release-specific statistics.

S05 Python 3 tutorial: introduction and control flow.

S10 DESeq2: Analyzing RNA-seq data with DESeq2, release vignette.

S20 Turkish Personal Data Protection Law No. 6698, official English translation with 2024 amendments.

S23 ICMJE recommendations: use of AI by authors.